

INDIA'S 5G STORY

While the world is on the verge of the next cellular network revolution, where does India stand?

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Since the last few months, after a deluge of announcements and promises at tech conferences and exhibitions, the world is anticipating 5G and the numerous benefits that it has been designed to bring. However, when it comes to India, the story is not that simple. Don't get us wrong, we are gearing up for 5G connectivity in our country as well, with multiple providers announcing their plans and strategies. However, once you dig a little deeper, and look at the specifics of this plan as well as the condition of the already existing 4G connectivity, things are not as straightforward.

WHY DO WE NEED 5G?

The overall benefits of 5G are obvious. Additionally, when something is cutting edge, there's always an interest in getting it into place as soon as possible. However, the benefits of something like 5G need to be analysed at a country-specific level especially for a country like India, where unique use cases have either given rise to new requirements altogether or done away with established, successful products that might have proven their mettle elsewhere considerably well.

5G is no different. While many of its global benefits are also suitable to India, some of its features are particularly apt for the country's requirements. For starters, the lower levels of investment required in physical infrastructure make it beneficial for India. In fact, this might lead to a spike in adoption levels, just like how telephone penetration in the country went from less than 5% in 1995 to 95% by 2010, and the economy benefited alongside, to the point where today, India offers the cheapest mobile data rates in the world. It would be no surprise if we see a similar spike in last mile connectivity with 5G.

WHAT ARE WE DOING?

The Government of India set up a high-level forum for 5G back in September 2017. The goal of this forum was to determine India's vision for 5G through the recommendation of policy changes as well as action plans. Last year, the Steering Committee of this forum prepared and released a report titled 'Making India 5G Ready' which includes a series of recommendations for various aspects of the roadmap for 5G in India.

After studying the different aspects of the 5G standards like eMBB (enhanced Mobile Broadband), mMTC (massive machine type communication)

and uRLLC (Ultra-reliable and Low Latency Communications), the committee made a number of recommendations.

The spectrum release for 5G has been recommended in three tiers, with respect to certainty:

- **Study tier:** For exploratory studies regarding 5G use.
- **Identify tier:** Bands designated for potential 5G use, after coordination with domestic users
- **Announce tier:** Bands announced for rollout

From an allocation point of view, the spectrum has been recommended for segmentation as follows:

- Radio Access Spectrum
- Radio Backhaul Spectrum
- 5 GHz Wi-Fi Spectrum

Radio Access Spectrum

The Radio Access Spectrum has been recommended for further subdivision as follows:

- The sub-1 GHz band with limited spectrum but suitable coverage has been recommended for mMTC as well as eMBB.
- The 1-6 GHz band with wider spectrum but reduced coverage has been recommended for eMBB and uRLLC applications.
- Millimetric bands in 21 to 41 GHz with enough spectrum but poor coverage have been recommended